Department of Data Science and Artificial Intelligence
National Institute of Technology Kurukshetra
B.Tech (AI&ML), AIPC 205/DBMS, Mid Term (I) Exam, ACY. 2025-26

Time Allowed: 50 Mins
Max. Marks: 15.00

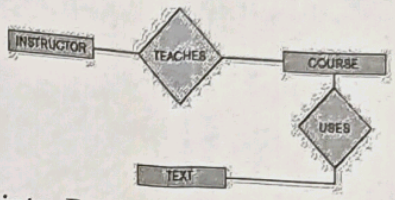
Date of Exam: 26.09.2025
Roll No: 124108055

Note: All the questions and their parts are compulsory; there is no choice. Answer each part of a question in the same place.

Q1. Anwar the following **briefly**: 1*4

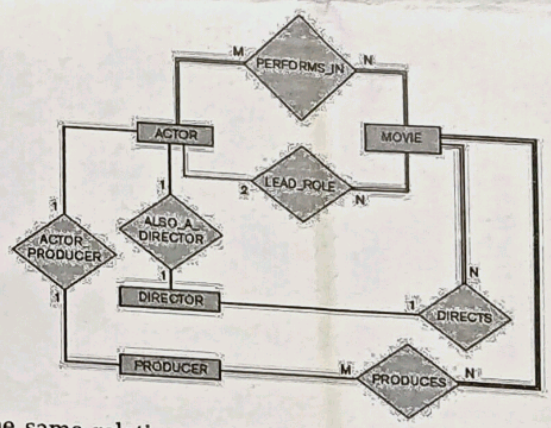
(a). Why data Model is important? How are the concepts of *Schema* related to each other?
(b). How is data retrieval different from information retrieval?
(c). Is it possible to successfully map a binary *M:N relationship type* without requiring a new relation? Why or why not?
(d). What is the *'Property of Uniqueness'* in a Relational DB, and how is it enforced?

Q2. (a) Consider the below *ER diagram*. Assume that a course must use a textbook. A course may not use more than *ten* books. Instructors teach from *two to three* courses. Supply *(min, max)* constraints on this diagram. State clearly any additional assumptions you make. If we add the relationship *ADOPTS*, to indicate the textbook(s) that an instructor uses for a course, should it be a binary relationship between *INSTRUCTOR* and *TEXT*, or a ternary relationship between all three entity types? ***What (min, max) constraints would you put on it? Justify?*** 2.00



(b) Consider the *MOVIES DB* schema, with set of constraints. Respond to the following statements with *True/ False/Maybe*. Assign a response of *Maybe* to statements that, while not explicitly shown to be *True*, cannot be proven *False* on the schema. Justify each answer: 3.00

(i) A movie can have only a maximum of two lead actors.
(ii) There are no actors in this DB that have been in no movies.
(iii) Most movies have one director and two producers.
(iv) No movie has a director who also acted in that movie.
(v) There are some actors who have done a lead role, directed a movie, and produced some movie.



Q3. *(a)* SQL allows a *foreign key* dependency to refer to the same relation, as in the following example: 2.00
Create table manager (employee_name varchar (20) primary key, manager_name varachar (20) not null, foreign key (manager_name) references manager (employee_name) on delete cascade on update set null);
Here, *employee_name* is a primary key to the table, meaning that each employee has at most one manager. The foreign key clause requires that every manager also be an employee. Explain exactly what will happen when a tuple in the relation *manager* is (i) deleted and (ii) updated.

(b) The National Election Management System is designed to handle the complex processes of electoral administration in a structured manner. Citizens are registered as voters and uniquely identified by a voter ID. Each voter belongs to a specific constituency, and constituencies are organized under states. Candidates contest elections within constituencies, with the condition that every candidate must also be a registered voter. Votes are recorded electronically, and each vote is associated with the voter, the candidate, and the constituency. Political parties play a crucial role by nominating candidates, while also allowing for the possibility of candidates switching parties over time. Campaign events are organized, where multiple candidates interact with and address voters. A recursive relationship exists where candidates may endorse or support other candidates. Furthermore, voters can be classified under specialization as Resident voters or Non-Resident Indians (NRIs), reflecting different participation categories in the electoral process. **Develop an ERD.** 4.00